



NATIONAL SCIENCE FOUNDATION
OFFICE OF SCIENCE AND TECHNOLOGY POLICY

Re: Request for Information on the Development of an Artificial Intelligence Action Plan

SUBMITTED MARCH 15, 2025
BY
CURT LEVEY
PRESIDENT
THE COMMITTEE FOR JUSTICE

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Introduction

The Committee for Justice (CFJ) appreciates the opportunity to respond to this important Request for Information regarding the development of an AI Action Plan pursuant to President Trump's Executive Order. We strongly support the Trump Administration's objective of maintaining America's continued leadership and dominance in artificial intelligence and believe the policy decisions incorporated into the Action Plan can help to ensure that the United States is the winner of the intense global AI competition.

CFJ favors pro-innovation AI policies that promote competition and realize AI's transformative potential, while discouraging burdensome litigation and avoiding unnecessary regulation that would slow technological progress and growth in America's AI industry. While furtherance of these goals implicates a wide range of issues, we focus these comments on the policies implicated by the recent series of lawsuits alleging that the developers of generative AI models have infringed the copyrights on material used to train these models.¹

These comments consist of recommendations for 1) regulations and policies that can be directly implemented by the Trump Administration, 2) Congressional action on copyright

¹ Generative AI models are ones that produce complex output patterns—typically images, audio, video, or text—after being trained on datasets consisting of materials of the same data type.

and related legislation that the Administration should encourage, and 3) judicial interpretations of existing law that the Administration should advocate for in legal memoranda and in court documents—including amicus briefs—as courts struggle with the legal issues presented by recent litigation concerning generative AI.

In sum, we recommend that the Administration’s Action Plan encourage the treatment of copyright-protected material used in AI training as fair use rather than copyright infringement, while focusing protection for copyright holders on the outputs of generative AI models, asking whether the outputs are substantially similar to the copyrighted training data.

Committee For Justice

Founded in 2002, CFJ is a nonprofit legal and policy organization that promotes and educates the public and policymakers about the rule of law and the benefits of constitutionally limited government. As part of this mission, CFJ advocates in Congress, the courts, and the news media about a variety of law and technology issues, involving intellectual property, antitrust law and competition policy, administrative law and regulatory reform, artificial intelligence, free speech, data privacy, and the impact of all these on innovation and economic growth.

Curt Levey, the author of these comments, is an attorney with a previous career as an artificial intelligence scientist, as well as legal and policy experience in intellectual property, including copyright. Prior to attending Harvard Law School, Mr. Levey studied AI at Brown University, earning undergraduate and Master’s degrees in computer science. He then joined HNC Software, an AI startup company, where he worked for five years as a staff scientist and invented and implemented a patented technology that provides explanations and confidence measures for the decisions made by neural networks.² The technology saw worldwide use in an AI system that detects payment-card fraud. More recently, Mr. Levey has written about AI policy³ and organized and moderated⁴, as well as participated in⁵, panels on the problem of bias in AI. On May 2,

² <https://patents.justia.com/patent/5398300>.

³ See, e.g., <https://www.wsj.com/articles/algorithms-with-minds-of-their-own-1510521093> (2017); <https://thehill.com/opinion/technology/432093-american-artificial-intelligence-strategy-offers-promising-start> (2019); <https://thehill.com/opinion/technology/444568-congress-can-bring-the-government-into-the-age-of-artificial-intelligence> (2019); <https://thehill.com/opinion/4563767-department-of-justice-is-best-choice-to-police-ai-competition> (2024).

⁴ <https://fedsoc.org/events/artificial-intelligence-and-bias> (2021)

2023, he testified at the U.S. Copyright Office’s listening session on the use of artificial intelligence to generate works of visual art⁶.

Comments

Issues surrounding the use of copyrighted material to train generative AI models have been at the forefront of legal and public debate over the past few years due to the filing of lawsuits claiming that the developers of generative AI models—including image generators and large language models—have infringed the copyrights on material contained in the training datasets. These datasets are used (some say “ingested”) by the models as part of the machine learning process.

The complaint in *Getty Images v. Stability AI*, which targets the Stable Diffusion model, is typical.⁷ It alleges infringement based both on the use of copyrighted images to train a generative AI model and on the possibility of that model generating images highly similar to and derivative of copyrighted images used in training.

The training materials that are used to build generative AI models and are at issue in these lawsuits come primarily from internet-crawled, publicly available material.⁸ To the extent that AI model builders want to use curated datasets or any other data that is not publicly available, their only lawful option is to contract for access to the data, and thus copyright infringement is not typically an issue.

In the short term, courts will have to resolve the issue of infringement by use of copyrighted material in training. In the longer term, this Administration’s Action Plan should urge Congress to step in and clarify the law—presumably by amending the Copyright Act of 1976—like it has done before when new technology has resulted in difficult new issues under copyright law. The approach to generative AI and copyright protection recommended in these comments should guide any statutory changes enacted by Congress, but it is also relevant to federal regulators and the courts, as they resolve novel legal issues.

⁵ <https://regproject.org/event/live-podcast-is-artificial-intelligence-biased-and-what-should-we-do-about-it> (2020); <https://fedsoc.org/events/artificial-intelligence-anti-discrimination-bias> (2023); <https://fedsoc.org/events/is-ai-woke-in-what-ways-and-should-we-worry> (2024).

⁶ <https://www.copyright.gov/ai/agenda/2023-Visual-Arts-Agenda.pdf>.

⁷ 1:23-cv-00135, (D. Del. filed 2023).

⁸ See <https://www.brookings.edu/articles/the-politics-of-ai-chatgpt-and-political-bias>.

A fair use regime is vital to American AI leadership

Maintaining American leadership in AI requires building the world's leading generative AI models, such as ChatGPT. Training these frontier models, in turn, requires access to huge amounts of data from the broadest possible range of sources, including in large part publicly available, copyrighted material. That access would be threatened should American courts and lawmakers, in response to lawsuits, decide that the use of this material in training constitutes copyright infringement rather than fair use.

Given such a regime, the largest players in the AI industry could afford to license some fraction of copyrighted training material. However, the practical effect would be to substantially limit the mountain of data that has made American leadership in AI possible and to make what access remains unpredictable, subject to lengthy negotiations, and generally more complicated. This would create steep barriers to entry that would make it difficult for smaller, newer innovators to compete with the AI giants that have much larger budgets and many more lawyers.

What the Action Plan should strive for is a balanced copyright regime that allows American AI models to learn from all of mankind's accumulated knowledge and creation, while protecting creators from the unauthorized duplication of their works. Fortunately, the fair use doctrine, a longstanding part of American copyright law, allows for transformative uses of copyrighted works. Generative AI models learn to produce new and different works—whether art, music, video, text, or the like—from examples and would thus seem to be the very definition of transformative. But courts and policymakers have yet to decide the question. Among the most important objectives of this Administration's AI Action Plan should be definitive resolution of this question in favor of fair use.

The alternative is to allow AI companies based in nations, such as China, that do not respect the protection of intellectual property, to gain an unfair advantage over American companies. As a practical matter, a Chinese company training a frontier AI model will have access to virtually all the world's data, copyrighted or not. However, if American courts and lawmakers rule against fair use access to copyrighted training data, AI companies in the United States will have only limited use of such data, making it difficult for them to compete on a playing field where data is king. Moreover, if the intent of the courts and lawmakers was to give priority to protecting creators, that intention will be largely thwarted as China and similar nations make unauthorized use of copyrighted training material to produce the leading generative AI models that creators are worried about.

Of course, the Administration's Action Plan should also include aggressive efforts to convince the nations of the world to adopt and enforce a balanced copyright regime similar to our own and to avoid putting obstacles in the way of America companies' use of publically available foreign data for training. However, that is easier said than done.

Human versus machine learning

Whether we are talking about the courts, which must be guided by existing statutes and precedent, or Congress, the good news is that there is no need to reinvent the wheel when approaching the question of copyright infringement by generative AI models. The similarities between machine learning in generative AI models and the manner in which human creators learn from a lifetime of examples makes it possible to apply the well-settled law governing the latter to the former with a result that will guarantee the robust access to training data necessary for American competitiveness. The Trump Administration's new AI Action Plan should encourage this approach.

In order to understand the similarities, consider that human creators learn a skill—painting or writing music, for example—from exposure to countless examples of art, music, and the like. The creator is not born with that skill. As a human learns from each new example, the synaptic strengths between neurons in their brains are slightly modified to reflect the new learning.

Similarly, neural networks, the technology underlying generative AI models, learn to generate images, music, and the like by being presented with a very large number of examples. Neural networks consist of analogues to biological neurons and synapses, typically simulated in software. As with humans, learning takes place as the synaptic strengths ("weights") are slowly modified in response to training examples.

It is well-settled law that a human learning a skill from the ingestion of publically available, copyrighted material does not constitute copyright infringement, even if that human goes on to compete with the creators of the original material without merely duplicating their work. OpenAI refers to that as the "freedom to learn" from copyrighted material and we see no statutory, policy, or logical reasons to deny that same freedom to the builders of AI models.⁹ If a human is free to learn about, say, painting from viewing copyrighted art works, does it make sense to say that a generative AI models cannot do the same?

⁹ See <https://openai.com/global-affairs/openai-proposals-for-the-us-ai-action-plan>.

There seems to be a common misconception that a trained generative AI model retains copies, in some form, of the individual training material. Take, for example, the complaint in *Andersen v. Stability AI*, alleging copyright infringement by popular image-generating AI models.¹⁰ The attorneys for the plaintiffs, who also represent plaintiffs in several of the other leading legal challenges to generative AI, assert that “By training Stable Diffusion on the Training Images, Stability caused those images to be stored at and incorporated into Stable Diffusion as compressed copies.”

In actuality, a trained AI model retains no copies of the training material. Retaining specific examples would be at odds with the objective of machine learning, which is to extract patterns and *generalize* from training data, rather than remember it. In fact, it is this aspect of neural networks, learning subtle relationships and patterns and encoding that knowledge in the interplay of the model’s weights, that makes the behavior of AI models so hard to control and explain.

Again, this is similar to how humans learn. While humans are capable of remembering a limited number of the specific examples they learn from, any sort of deep learning—whether it involves learning a creative skill or just recognizing the difference between a dog and a wolf—requires generalization from examples rather than detailed storage of them. When dealing with copyright issues, the AI Action Plan should be mindful of the similarities between machine and human learning.

The focus of copyright protection should be on AI models’ outputs

When a human learns from ingesting examples in the domain they are trying to master, much of that material is typically protected by copyright. Yet that ingestion goes unchallenged, being regarded as either fair use or not subject to the Copyright Act at all. However, if that human uses their resulting learning to duplicate or produce a derivative work from one of the copyrighted examples without authorization, they are liable for copyright infringement.

To put it another way, when assessing copyright infringement by humans, the law looks at the potentially infringing work the human produced rather than the process used to produce it, whether the process is the examples he learned the relevant skills from or the tools he used to produce the work.

We suggest the new AI Action Plan encourage a similar approach to assessing copyright infringement by generative AI models—that is, a focus on the outputs of the

¹⁰ 700 F. Supp. 3d 853 (N.D. Cal. 2023).

model. Where an output is “substantially similar” to any of the copyright-protected training material—or any copyrighted work for that matter—it should be treated as a derivative work, subject to liability for infringement.¹¹

Moreover, while the mere use of copyrighted data in a training dataset should be presumed to be fair use, it should be this Administration’s policy that the presumption can be rebutted by showing that there was copying and retention of copyrighted training material beyond that necessary to train the AI model (more on this later).

Reasons to rely on human-based policies

This is a good point at which to explain why the AI Action Plan should rely on analogies to human learning and creation in determining how to apply copyright law to potential infringement by generative AI. One reason is that analogizing allows policymakers to more easily adapt existing law when faced with new technology. Dealing with the many legal and policy challenges reflected in the AI and copyright debate is hard enough without reinventing the wheel. The Trump Administration should use the similarity between human and machine learning to their advantage to guide policy development.

Similar reasoning should govern the Administration’s reasoning when it weighs in on the legal issues presented by litigation concerning generative AI. Courts are required to apply existing law rather than creating new policies. So when faced with relatively novel legal issues resulting from new technology, courts rely on analogies. For example, while the Fourth Amendment’s guarantee of privacy in one’s “persons, houses, papers” does not apply directly to mobile phones and email accounts, the courts have analyzed privacy in these newer domains by analogizing to homes, documents, and the like. So too, this Administration should urge courts to analogize to human learning when addressing generative AI.

There is also much to be said for intellectual consistency. While it may philosophically trouble some people to recognize the similarities between human and machine learning, the burden should be on those who want to treat human and machine learning as incomparable phenomena for copyright purposes, despite the similarities. Considering that the distinction between human and machine cognition will surely narrow as AI technology progresses, Administration policies that downplay the similarities are unlikely to stand the test of time.

¹¹ The question of which person or entity should be liable—the developers of the model, the developers of the training dataset, the user whose prompt resulted in the infringing output, or someone else—is beyond the scope of these comments.

Finally, consider that the analogy between human and machine learning can work in both directions. Any Administration policies that treat the ingestion of copyrighted material for learning purposes as infringement when performed by a generative AI model may someday blur the lawfulness of the same use of copyrighted material for human learning, especially as technology narrows the distinction between the two.

Providing creators with additional protection against derivative works

While denying the use of copyrighted training material the protection of the fair use doctrine would be disastrous for American leadership in AI, the Action Plan should provide balance by looking for ways to protect the rights of creators. Even putting aside the human analogy, the approach taken by recent lawsuits—that is, defining the use of copyrighted material to train generative AI models as copyright infringement—is unlikely to be a robust solution to protecting the rights of creators. But before discussing this in more detail in the next section, we discuss how the alternative approach—that is, protecting creators' copyrights by focusing on a model's outputs—can be strengthened by policymakers. Specifically, the AI Action Plan should encourage Congress to make two changes to the Copyright Act.

Congress should specify that a finding that a derivative work is transformative, which weighs heavily in favor of finding that the work is fair use, faces a higher bar when the work is produced by generative AI, unless the potential infringer can show that the transformative quality is the product of a human's prompts. This modification to existing law makes sense when we consider that a transformative work is one with “a further purpose or different character” than the original work, as the Supreme Court explained in *Andy Warhol Foundation for the Visual Arts v. Goldsmith*.¹²

Determining the “purpose” of a derivative work presumes intent on the part of its creator. So too for a “different character” that is intentional. As impressive as generative AI models are, it is not claimed that they have anything akin to human intent with regard to the nature of the outputs they produce. Therefore, these models' outputs should be subject to a higher standard to meet the definition of a transformative work.

A second statutory change Congress should make to protect creators is to specify that the threshold of “substantial similarity” necessary to find that a work is derivative should be lower when that work is produced by generative AI, if there is evidence that the original work was in the training dataset. This makes sense because of the possibility

¹² 143 S. Ct. 1258 (2023).

that the presence of the original work in the training data indirectly contributed—by influencing the relationships learned by the neural network—to the substantial similarity, rather than it being a virtual coincidence.

Focusing on training data is not a robust solution for protecting creators' rights

Recall that machine learning, like human learning, does not depend on the retention of material in the training dataset. Once that data is used to train the weights of the neural network, it can be discarded without any degradation in the performance of a generative AI model. While more *permanent* storage of training material is common—whether by AI model builders or dataset curators—and can be an independent basis for copyright infringement, this presents a separate issue from whether the use of copyrighted material merely for training is, in and of itself, copyright infringement. For training purposes, *temporary* copying is all that is fundamentally necessary.

In pursuit of protecting creators' copyrights, plaintiffs argue that even temporary copying by the developers of AI models is infringement. But that is a weak peg to hang this Administration's policies on. For one thing, it is not fundamentally different from what humans do when they learn from examples. For convenience sake, humans often intentionally make copies of the material—music, art, articles—that they use as learning examples. And even when they don't, their computers make temporary copies of the songs, images, and text they use to master a skill. Yet all of that copying is generally accepted as fair use.

As the Electronic Frontier Foundation has pointed out:

“‘Temporary copying’ of data is fundamental to how computing works in general, especially on the Internet. For example, ... browser cache files are stored on servers to speed up the loading of websites, and copies of visited pages are stored in a temporary Internet files folder on your hard drive, speeding up the loading process for those websites the next time you visit them.”¹³

In recognition of this fact, as EFF notes, U.S. courts have generally held that temporary copying is either not subject to the Copyright Act or is fair use.

It is worth noting that developing a generative AI model doesn't truly require even a temporarily copied training dataset. If copying made the developers liable for infringement, they could instead construct a training process that consisted of scrolling through the publicly available material stored on the internet, rather than gathering that

¹³ https://www.eff.org/files/filenode/temporary_copies_fnl.pdf.

material into a dataset. While it would make for a slower training process, the point is that meaningful copying is not fundamentally necessary for training. When looking for ways to protect creators' rights, the AI Action Plan would be advised to look elsewhere.

Another reason why a focus on the use of copyrighted materials for training is a misguided policy strategy is that it is not aimed at the real threat to copyright holders. Consider that prior to the emergence of generative AI, more simple neural network models—typically classification and scoring models—were trained on large amounts of data (numerical, visual, acoustic, textual, and the like), some portion of which was copyrighted. Yet there was little or no objection from the copyright holders. The recent explosion in protests and lawsuits by copyright holders is motivated largely by the fear that the outputs of generative AI will negatively affect the potential market for their copyrighted works.

While that is a legitimate, if speculative, concern and is a factor in fair use analysis, it is a concern focused entirely on the outputs of generative AI. Competition from generative AI would be a concern to human creators whether or not their works were used to train AI models. To the extent that the protection of creators' rights in the AI Action Plan focuses on training data, it is flailing at a peripheral issue and distracting policymakers and the courts from the actual competitive threat posed by the outputs of generative AI.